CYART SOC TEAM – WEEK 4 REPORT

Name: Tanay Sengupta
Role: SOC Intern
Organization: CyArt Tech LLP
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Executive Summary

During Week-4 of the SOC Internship program, I performed advanced Security Operations Center (SOC) activities focusing on threat hunting methodologies, SOAR automation, adversary emulation, alert triage automation, incident response workflow simulation, and executive-level reporting.

The objective of this week's assignment was to strengthen proactive detection capabilities, automate repetitive SOC workflows, simulate attacker techniques using MITRE ATT&CK mappings, and perform post-incident analysis with measurable SOC performance metrics such as MTTD and MTTR.

Multiple industry-standard tools were used including:

- Elastic Security
- Wazuh
- Velociraptor
- AlienVault OTX
- MITRE Caldera
- TheHive
- CrowdSec
- Metasploit

This exercise simulated real SOC analyst responsibilities including investigation, automation design, detection validation, response containment, and reporting to stakeholders.

1. Threat Hunting Methodologies

Objective

To develop hypothesis-driven threat hunting skills using structured frameworks and log analysis techniques mapped to MITRE ATT&CK.

Approach

A threat hunting hypothesis was created:

> Unauthorized privilege escalation may occur within domain accounts.

Elastic Security logs were queried for:

Event ID: 4672 (Special privileges assigned to new logon)

Example findings:

Timestamp	User	Event ID	Notes
2025-08-18 15:00	testuser	4672	Unexpected admin role detected

The alert was mapped to:

MITRE ATT&CK Technique: T1078 – Valid Accounts

Threat intelligence indicators were further validated using:

- AlienVault OTX
- Velociraptor process queries

Result:

No confirmed compromise indicators were detected, but anomaly behavior validated monitoring rules effectiveness.

2. SOAR Playbook Development

Objective

Automate phishing response workflow using orchestration tools.

Tools Used

- Splunk Phantom
- CrowdSec
- TheHive
- Wazuh

Playbook Workflow

Steps executed:

1. Detect phishing alert from Wazuh
2. Extract suspicious IP address
3. Perform reputation lookup
4. Block malicious IP using CrowdSec
5. Automatically generate incident ticket in TheHive

Example execution log:

Playbook Step	Status	Notes
IP reputation check	Success	IP flagged malicious
IP block action	Success	Blocked attacker IP
Case creation	Success	TheHive ticket generated

Result:

Automation reduced response time significantly compared to manual investigation.

3. Post-Incident Analysis (Root Cause Analysis)

Objective

Identify cause of simulated phishing attack using structured RCA techniques.

5-Whys Analysis

Question	Answer
Why was email opened?	User clicked malicious link
Why was link clicked?	Email appeared legitimate
Why not blocked earlier?	Weak email filtering
Why filtering weak?	Missing IOC rules
Why rules missing?	Incomplete threat intelligence integration

Root Cause:

Insufficient phishing detection filtering rules.

Fishbone Diagram Observations

Incident causes categorized into:

- Technology gaps
- Detection limitations
- User awareness issues
- Email filtering misconfiguration

Result:

Recommendations were prepared for improving detection pipelines.

SOC Metrics Calculation

Example values:

Metric	Value
MTTD	2 hours
MTTR	4 hours

Interpretation:

Detection efficiency acceptable but response automation improvements recommended.

4. Alert Triage with Automation

Objective

Perform automated alert validation using threat intelligence integration.

Alert Example

Alert ID	Description	Source IP	Priority	Status
005	Suspicious File Download	192.168.1.102	High	Open

Validation Steps:

1. Alert investigated in Wazuh dashboard
2. File hash extracted
3. Hash checked via VirusTotal integration inside TheHive
4. Threat severity confirmed automatically

Result:

Automation accelerated investigation workflow and reduced manual workload.

5. Evidence Analysis

Objective

Analyze endpoint network activity using Velociraptor.

Command executed:

SELECT * FROM netstat

Findings:

Suspicious outbound connections identified and verified against threat intelligence feeds.

Chain of Custody Documentation

Item	Description	Collected By	Date	Hash
Network Log	Server-Z Log	SOC Analyst	2025-08-18	SHA256 Generated

Maintaining evidence integrity ensures forensic reliability.

6. Adversary Emulation Practice

Objective

Simulate attacker behavior using MITRE Caldera.

Technique simulated:

T1566 – Phishing Attack

Detection workflow:

Timestamp	TTP	Detection Status	Notes
2025-08-18 17:00	T1566	Detected	Email blocked successfully

Result:

Wazuh detection rules successfully identified phishing indicators.

Detection gap:

Minor delay observed in alert correlation timing.

7. Security Metrics Dashboard & Executive Reporting

Metrics Generated

Dashboard created using Elastic Security containing:

- Mean Time to Detect
- Mean Time to Respond
- False Positive Rate
- Incident Resolution Rate
- Dwell Time

Example metrics:

Metric	Value
MTTD	2 hours
MTTR	4 hours
False Positive Rate	Low

Interpretation:

SOC detection pipeline functioning effectively but automation expansion recommended.

8. Capstone Project – Full SOC Incident Response Simulation

Objective

Simulate real-world cyberattack lifecycle and perform full investigation workflow.

Attack Simulation

Tool used:Metasploit

Exploit executed:exploit/multi/samba/usermap_script

Target:Metasploitable2 VM

Result:Remote service exploitation successful.

Mapped Technique:MITRE ATT&CK T1210 – Exploitation of Remote Services

Detection Phase

Wazuh generated alert:

Timestamp	Source IP	Alert	Technique
2025-08-18 16:00	192.168.1.102	Samba exploit detected	T1210

Triage Phase

Incident analyzed in:

TheHive

Actions performed:

- Alert classification
- Severity assignment
- IOC extraction
- Timeline creation

Response & Containment Phase

Containment actions executed:

- Target VM isolated
- Attacker IP blocked using CrowdSec
- Connectivity verification performed

Result:

Attack successfully contained.

SOAR Automation Integration

Automated workflow executed:

1. Detection triggered
2. Case created automatically
3. IP blocked
4. Analyst notified

Automation improved response efficiency.

Post-Incident Root Cause Analysis

Using 5-Whys technique:

Root cause identified:

Unpatched Samba service vulnerability exposed to network access.

Metrics Reporting

Calculated:

Metric	Value
MTTD	2 hours
MTTR	4 hours
Dwell Time	Low

Dashboard created inside Elastic Security.

Stakeholder Briefing (Executive Version)

A simulated remote service exploitation attack was detected targeting the Metasploitable2 environment. The SOC successfully identified suspicious activity using Wazuh alerts mapped to MITRE ATT&CK technique T1210. Automated response playbooks blocked the attacker's IP and isolated the compromised system. Root cause analysis revealed an unpatched Samba vulnerability. Recommendations include vulnerability patching, automated alert correlation improvements, and enhanced phishing awareness training. Overall SOC response effectiveness remained strong with low dwell time and acceptable detection speed.

Conclusion

This week's SOC activities provided hands-on exposure to:

Threat hunting
SOAR automation
Alert triage workflows
Evidence handling
Adversary emulation
Incident response lifecycle
Security metrics reporting
Executive-level communication

The simulated attack scenarios strengthened practical detection and response capabilities aligned with modern SOC operational standards and MITRE ATT&CK methodology.